

Capturing traffic over Wi-Fi

Upload media immediately

Privacy Analysis of Augmented Reality Glasses

1 Introduction

This experiment captures network traffic over Wi-Fi during the interaction between an Android phone and the Ray Ban Meta glasses, more specifically while executing touch commands. Note that the photo/videos taken are manually imported onto the phone immediately after the capture. The aim of this data collection is to observe the network traffic generated during typical usage of the glasses (to see how data flows between the device, its companion app and other apps). The goal would be to identify potential vulnerabilities or privacy breaches if data transmission is not securely protected. Furthermore, we can take a peek into whether any personally identifiable information or metadata can be taken during daily usage, therefore observe how user privacy may be affected.

2 Metadata

Time of experiment: Oct 16th 00:05

Duration of experiment: Roughly 13 minutes

IP addresses:

- Laptop - 192.168.0.215
- Redmi phone - 192.168.1.141
- Wi-Fi Network - 192.168.0.1
- ALFA Hotspot - 192.168.1.100/24
- Meta CDN (content delivery network) - 157.240.17.17

3 Touch Commands used

- Keep idle for 1 minute (glasses off)
- Open Meta View
- Power On
- Pairing device
- Short click capture button to take a photo
- Upload image to phone
- Long click capture button to take a video recording
- Short click capture button to stop video recording
- Upload video to phone
- Long press capture button to take a full length video recording
- Upload video to phone
- Tap (and hold) on the side to play music

- Swipe forward to increase the volume
- Swipe backwards to decrease the volume
- Double tap to skip
- Triple tap to go previous
- Tap on the side to stop music
- Keep idle for 5 minutes (glasses on)

TODO:

4 Time Stamps

5 Note